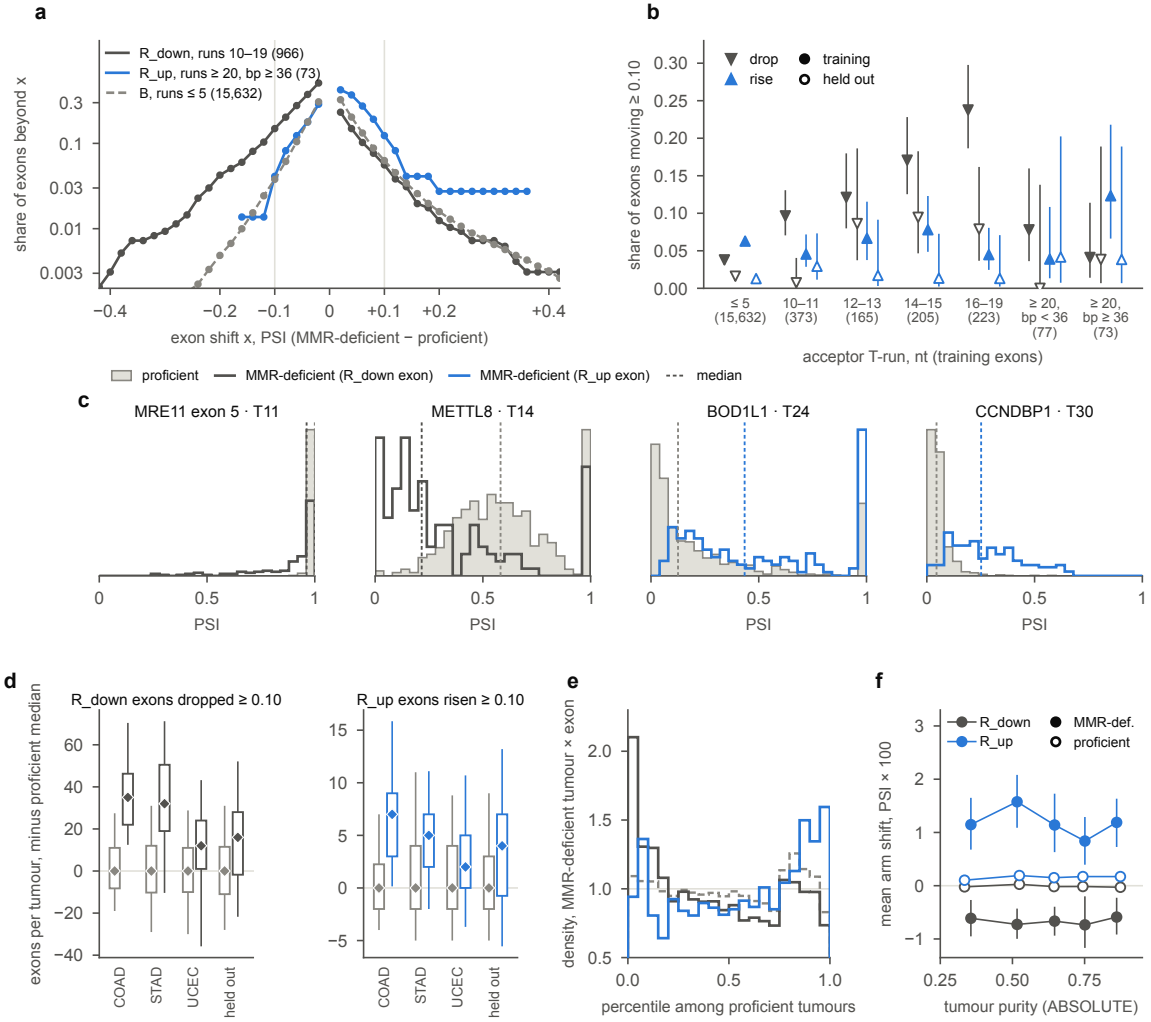
a
R_down, runs 10–19 (966)
R_up, runs ≥ 20, bp ≥ 36 (73)
B, runs ≤ 5 (15,632)
share of exons beyond x
0.3
0.1
0.03
0.01
0.003
−0.4
−0.2
−0.1
0
+0.1
+0.2
+0.4
exon shift x, PSI (MMR-deficient − proficient)
b
drop
rise
training
held out
share of exons moving ≥ 0.10
0.30
0.25
0.20
0.15
0.10
0.05
0.00
≤ 5 (15,632)
10–11 (373)
12–13 (165)
14–15 (205)
16–19 (223)
≥ 20, bp < 36 (77)
≥ 20, bp ≥ 36 (73)
acceptor T-run, nt (training exons)
proficient
MMR-deficient (R_down exon)
MMR-deficient (R_up exon)
median
c
MRE11 exon 5 · T11
METTL8 · T14
BOD1L1 · T24
CCNDBP1 · T30
0
0.5
1
PSI
d
R_down exons dropped ≥ 0.10
R_up exons risen ≥ 0.10
exons per tumour, minus proficient median
60
40
20
0
−20
−40
15
10
5
0
−5
COAD
STAD
UCEC
held out
e
density, MMR-deficient tumour × exon
2.0
1.5
1.0
0.5
0.0
0.5
1.0
percentile among proficient tumours
f
R_down
R_up
MMR-def.
proficient
mean arm shift, PSI × 100
3
2
1
0
−1
0.25
0.50
0.75
tumour purity (ABSOLUTE)